W12

Landscape metrics and final projects

GIS & MAPPING | R PROGRAMMING

Landscape metrics quantify composition (how much of each habitat) and configuration (how it is arranged: patch size, shape and connectivity). Today you use them to measure how the Sierra de Montfred changed between 1990 and 2026 with rural abandonment, and we close the course with your projects.

DURATION	1 hour of metrics and 1 hour of presentations.
PACKAGES	`landscapemetrics`, `terra`, `sf`, `tidyverse`
FILES	`week12_landscape_metrics.R`, `week12_solutions.R`, `landcover.tif`, `landcover1990.tif`, `landcover_classes.csv`, `montfred.gpkg`
DATA	Land cover in 1990 and 2026 (simulated) inside the park.

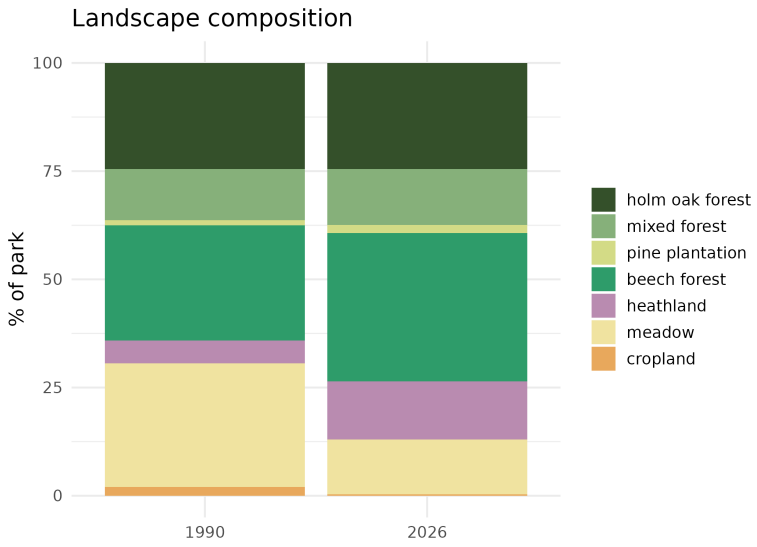
Learning outcomes

- tell landscape composition from configuration;
- calculate patch, class and landscape metrics with landscapemetrics;
- build a transition matrix between two dates;
- understand that metrics depend on scale and resolution;
- present a complete spatial analysis, from data to map.

1 · Composition: what changed

landscapemetrics functions are named `lsm_` + level + metric: `lsm_c_pland()` is the percentage of each class, `lsm_l_shdi()` the landscape's Shannon diversity.

Class	1990 (%)	2026 (%)	Change
meadow	28.6	12.7	-15.9
beech forest	26.5	34.3	+7.8
heathland	5.3	13.4	+8.1
cropland	2.1	0.3	-1.8
mixed forest	11.9	12.9	+1.0
holm oak forest	24.5	24.5	0



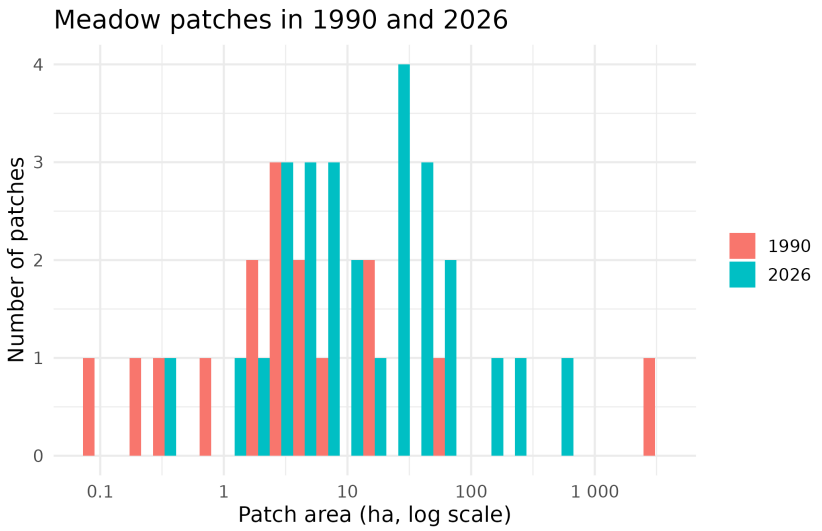
Park composition in 1990 and 2026.

The transition matrix (`crosstab()`) shows what each cell became: of about 3,200 ha of meadow in 1990, about 910 ha are now heathland and about 870 ha beech forest. This is the typical pattern in Mediterranean mountains after grazing is abandoned: scrub and forest close in on the pastures.

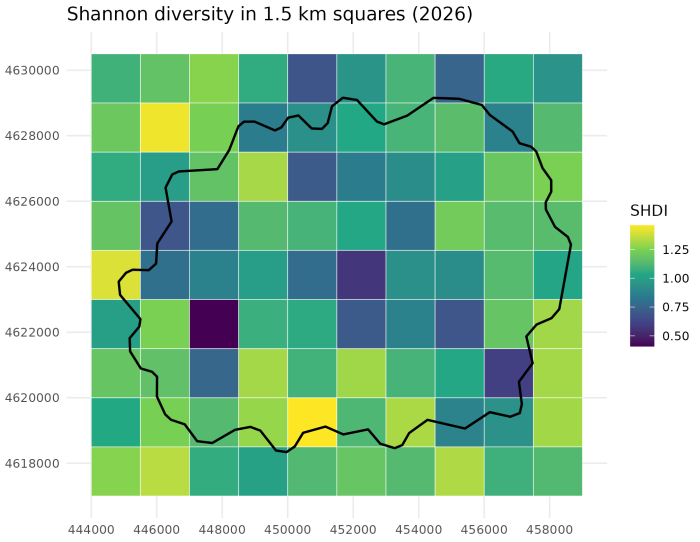
2 · Configuration: fragmentation

Metric	What it measures	1990	2026
np	number of patches (all classes)	296	215
ed	edge density (m/ha)	41.1	38.0
lpi	% covered by the largest patch	27.5	32.1
shdi	Shannon diversity	1.60	1.60
meadow: patches		16	27
meadow: mean area (ha)		200	52

At landscape level there are fewer patches and less edge: forest has consolidated and the landscape is more homogeneous, although Shannon diversity barely changes. But meadows have fragmented: more patches, four times smaller. For mountain-grassland species, that is what matters.



Meadow patch sizes in 1990 and 2026 (log scale).



Shannon diversity in 1.5 km squares (`sample_lsm()`): the landscape is most diverse in the foothills, at the park edge and beyond, where cropland, pine and holm oak mix; the forested interior is more uniform.

Metrics depend on scale

Changing raster resolution, extent or the neighbourhood rule (4 or 8 cells) changes the values. Only compare metrics calculated with the same settings, and choose a few that answer your ecological question: many are strongly correlated with each other.

3 · Final project

Each person (or pair) presents a short analysis using the course data or their own. Length: 5 minutes plus 2 for questions, 3–5 slides.

1. **Question:** one specific ecological question. "Where are woodlarks most abundant?" beats "I analyse the birds".
2. **Data:** where they come from, how many, what cleaning you did.
3. **Methods:** which analysis and why (for example, binomial GLM because the response is 0/1).
4. **Result:** one map and one figure or table, made in R or QGIS, readable from the back of the room.
5. **Conclusion and limits:** what it answers and what it does not; what you would do with more data.

Criterion	Weight
Clear question that fits the data	20%
Correct, justified methods	30%
Quality of map and figures	25%
Ecological interpretation and limits	15%
Reproducible script submitted	10%

Ideas with the course data: how will meadow loss affect woodlark habitat? (Weeks 10 and 12); which landscape variables within 250 m best explain richness? (Weeks 4 and 6); do the genets' core areas overlap? (Week 11).

Exercises

Answers in `week12_solutions.R`.

1. How many hectares of 1990 meadow are heathland today? And beech forest?
2. Beech forest edge density in 1990 and 2026.
3. Redo the diversity map with 3 km squares. Does the pattern change?
4. Pick three metrics we have not used (`list_lsm()`) and explain what they measure.

To keep learning

- *Geocomputation with R* (r.geocompx.org): the reference for sf and terra, free.
- *Spatial Data Science*, Pebesma and Bivand (r-spatial.org/book).
- QGIS Training Manual (docs.qgis.org).
- Fletcher and Fortin (2018), *Spatial Ecology and Conservation Modeling*, Springer: R examples of everything in this course.